## P0171-FUEL SYSTEM 1/1 LEAN — Continued

| TEST | ACTION | APPLICABILITY |
|---|---|---|
| 8 | **NOTE: For this test to be valid, the thermostat must be operating correctly.**<br>**NOTE: This test works best if performed on a cold engine (cold soak)**<br>Ignition on, engine not running.<br>With the DRBIII®, read the Engine Coolant Temperature (ECT) Sensor value. If the engine was allowed to sit overnight (cold soak), the temperature value should be a sensible value that is somewhere close to the ambient temperature.<br>**NOTE: If engine coolant temperature is above 82°C (180°F), allow the engine to cool until 65°C (150°F) is reached.**<br>Start the Engine.<br>During engine warm-up, monitor the ECT Sensor value. The temperature value change should be a smooth transition from start up to normal operating temperature 82°C (180°F). The value should reach at least 82°C (180°F).<br>Did the ECT value increase smoothly and did it reach at least 180°F (82°C)??<br>Yes → Go To 9<br>No → Replace the Engine Coolant Temperature Sensor.<br>Perform POWERTRAIN VERIFICATION TEST VER - 5. | All |
| 9 | Turn the ignition off.<br>Check for any of the following conditions/mechanical problems.<br>AIR INDUCTION SYSTEM - must be free from leaks.<br>ENGINE VACUUM - must be at least 13 inches in neutral<br>ENGINE VALVE TIMING - must be within specifications<br>ENGINE COMPRESSION - must be within specifications<br>ENGINE EXHAUST SYSTEM - must be free of any restrictions or leaks.<br>ENGINE PCV SYSTEM - must flow freely<br>TORQUE CONVERTER STALL SPEED - must be within specifications<br>POWER BRAKE BOOSTER - no internal vacuum leaks<br>FUEL - must be free of contamination<br>FUEL INJECTOR - plugged or restricted injector; control wire not connected to correct injector<br>Are there any engine mechanical problems?<br>Yes → Repair as necessary.<br>Perform POWERTRAIN VERIFICATION TEST VER - 5.<br>No → Go To 12 | All |
| 10 | Ignition on, engine not running.<br>**NOTE: Perform the following test on the O2 Sensors whose voltage was not switching properly in the previous step.**<br>Disconnect the O2 Sensor harness connector.<br>With the DRBIII®, monitor the O2 Sensor voltage.<br>O2 Sensor voltage should read 5.0 volts on the DRBIII® with the connector disconnected.<br>Connect a jumper wire between the O2 Signal circuit and the O2 Return circuit in the O2 Sensor harness connector.<br>**NOTE: The voltage should drop from 5.0 volts to 2.5 volts with the jumper wire in place.**<br>Did the O2 Sensor voltage change from 5.0 volts to 2.5 volts?<br>Yes → Replace the O2 Sensor.<br>Perform POWERTRAIN VERIFICATION TEST VER - 5.<br>No → Go To 11<br>**NOTE: Remove the jumper wire before continuing.** | All |